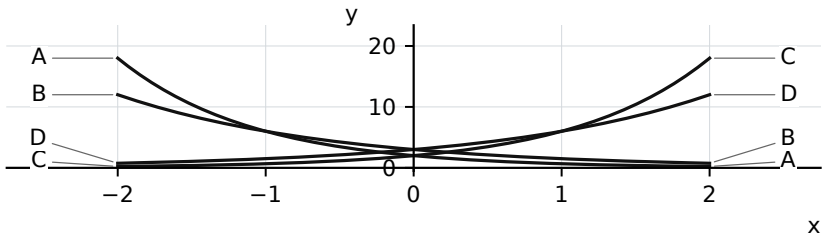
MATH 101 Final Exam

Practice 3 | Version 202610122020.48

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

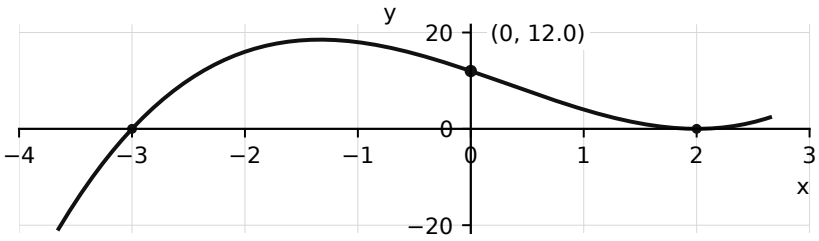
QUESTIONS	ANSWERS
1. Evaluate exactly using logarithmic properties; show the property used. Here log means base 10. (a) $\log(0.05) + \log(200000)$ (b) $\frac{\log_8(125)}{\log_8(25)}$	1. (a) 5 pts (b) 5 pts
2. Which pair satisfies h(x) = f(g(x))? $h(x) = (5x+5)^2 + 6$ $A.\quad f(x) = 5x^2 + 6,\ g(x) = x + 5$ $B.\quad f(x) = 5x + 5,\ g(x) = x^2 + 6$ $C.\quad f(x) = x^2 + 5,\ g(x) = 5x + 6$ $D.\quad f(x) = x^2 + 6,\ g(x) = 5x + 5$	2. Choice 10 pts
3. Match each function to graph A, B, C, or D. Each letter is used once. (a) $2(3)^x$ (b) $2(\frac{1}{3})^x$ (c) $3(2)^x$ (d) $3(\frac{1}{2})^x$ 	3. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts

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QUESTIONS	ANSWERS
4. Find (a) domain and range; (b) both intercepts; (c) behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. $f(x) = 3\sqrt{x+16} - 12$	4. (a) Domain 2 pts (a) Range 2 pts (b) x-intercept 1 pts (b) y-intercept 1 pts (c) Infinite end 3 pts (c) Finite end 1 pts
5. Find all zeros: (a) list Rational Root Theorem candidates, then test only -1 and 1; (b) use polynomial long division by the confirmed linear factor; (c) use the quadratic formula on the quotient. Record all zeros. $p(x) = x^3 + 3x^2 - 2$	5. (a) Candidates/root 2 pts (b) Quotient 4 pts (c) Other zeros 4 pts
6. Solve exactly. $8^{x-3} = 4^{x+4}$	6. Solution 10 pts
7. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	7. Polynomial 10 pts

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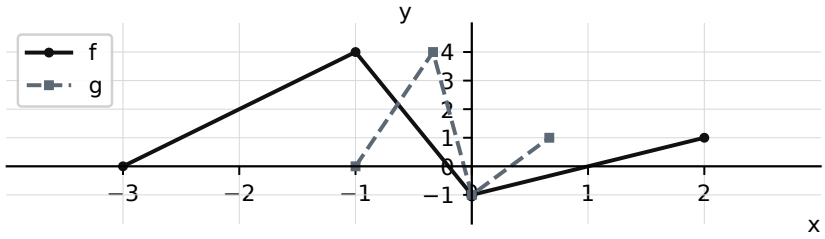
QUESTIONS	ANSWERS
8. Find (a) domain and range; (b) vertical asymptote and x-intercept; (c) behavior at both ends of the domain. $f(x) = \log_4(x - 4) + 3$	8. (a) Domain 2 pts (a) Range 2 pts (b) Asymptote 1 pts (b) x-intercept 1 pts (c) Left end 2 pts (c) Right end 2 pts
9. Find (a) domain; (b) hole, vertical asymptote, and horizontal asymptote; (c) both intercepts. $f(x) = \frac{(x+4)(x+5)}{(x-7)(x+4)}$	9. (a) Domain 2 pts (b) Hole 2 pts (b) Vertical 1 pts (b) Horizontal 1 pts (c) x-intercept 2 pts (c) y-intercept 2 pts

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QUESTIONS	ANSWERS
10. The solid graph is f and the dashed graph is g. Select the expression for g(x). *A*. $g(x) = f(3x)$ *B*. $g(x) = f(x - 3)$ *C*. $g(x) = f(x/3)$ *D*. $g(x) = 3f(x)$ 	10. Choice 10 pts

Instructor answer key: MATH 101 Final Exam

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a)
4
5 pts

(b)
$\frac{3}{2}$
5 pts

2. Choice
D
10 pts

3. (a)
C
3 pts

(b)
A
2 pts

(c)
D
3 pts

(d)
B
2 pts

TEST PAGE 2

4. (a) Domain
$[-16, \infty)$ 2 pts

(a) Range
$[-12, \infty)$ 2 pts

(b) x-intercept
$(0,\ 0)$ 1 pts

(b) y-intercept
$(0,\ 0)$ 1 pts

(c) Infinite end
$x \to \infty :\ f(x) \to \infty$ 3 pts

(c) Finite end
$x \to -16^{+} :\ f(x) \to -12$ 1 pts

5. (a) Candidates/root
$\{-2, -1, 1, 2\};\ x = -1$ 2 pts

(b) Quotient
$x^2 + 2x - 2;\ R = 0$ 4 pts

(c) Other zeros
$x = -1 + \sqrt{3},\ -\sqrt{3} - 1$ 4 pts

6. Solution
$x = 17$
10 pts

7. Polynomial
$1(x + 3)(x - 2)^2$
10 pts

TEST PAGE 3

8. (a) Domain
$(4, \infty)$ 2 pts

(a) Range
$(-\infty, \infty)$ 2 pts

(b) Asymptote
$x = 4$ 1 pts

(b) x-intercept
$(\frac{257}{64},\ 0)$ 1 pts

(c) Left end
$x \to 4^{+} :\ f(x) \to -\infty$ 2 pts

(c) Right end
$x \to \infty :\ f(x) \to \infty$ 2 pts

9. (a) Domain
$\mathbb{R} \setminus \{-4, 7\}$ 2 pts

(b) Hole
$(-4,\ -\frac{1}{11})$ 2 pts

(b) Vertical
$x = 7$ 1 pts

(b) Horizontal
$y = 1$ 1 pts

(c) x-intercept
$(-5,\ 0)$ 2 pts

(c) y-intercept
$(0,\ -\frac{5}{7})$ 2 pts

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TEST PAGE 4

10. Choice

A

10 pts